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ADBMS-Expt 5

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Experiment 05

Medium

(A) A website maintains records of users who have registered on the platform and the status of confirmation requests sent to them. A confirmation request can either be successfully completed or may expire without completion.
For each registered user, calculate the confirmation rate, which is defined as:

```
Confirmation Rate = Number of Successful Confirmations
                    -----------------------------------
                    Total Confirmation Requests
```

If a user has never received any confirmation request, their confirmation rate should be considered **0**.

Display the user ID along with the confirmation rate rounded to **2 decimal places**. The result can be returned in any order.

Example
Suppose a user has the following confirmation records:

Action
confirmed
confirmed
timeout

Then:

Successful Confirmations = 2
Total Requests = 3

Confirmation Rate = 2 / 3 = 0.67

Query   Query History

```
);
INSERT INTO Signups
VALUES
(3,'2020-03-21 10:16:13'),
(7,'2020-01-04 13:57:59'),
(2,'2020-07-29 23:09:44'),
(6,'2020-12-09 10:39:37');


INSERT INTO Confirmations
VALUES
(3,'2021-01-06 03:30:46','timeout'),
(3,'2021-07-14 14:00:00','timeout'),
(7,'2021-06-12 11:57:29','confirmed'),
(7,'2021-06-13 12:58:28','confirmed'),
(7,'2021-06-14 13:59:27','confirmed'),
(2,'2021-01-22 00:00:00','confirmed'),
(2,'2021-02-28 23:59:59','timeout');


select s.user_id, round(avg(case when c.action='confirmed' then 1 else 0 end),2) as confirmation_rate from Signups s
left join Confirmations c
on s.user_id=c.user_id
group by s.user_id
order by confirmation_rate desc
```

Data Output   Messages   Notifications

SQL   Showing rows: 1 to 4   Page No: 1 of 1

	user_id [PK] integer	confirmation_rate numeric
1	7	1.00
2	2	0.50
3	6	0.00
4	3	0.00

Experiment 05

Hard

(B) Problem Statement

A ride-booking company maintains information about trip requests and the users involved in those trips. Some users may be restricted from using the platform, while others are active participants. Each trip can either be completed successfully or cancelled by one of the participants.

For each day between **2013-10-01** and **2013-10-03**, determine the proportion of canceled trips among all valid trips. A trip should be considered valid only when both the customer and the driver associated with that trip are active users. The final result should display the date and the calculated cancellation rate rounded to two decimal places.

We need to calculate:

Cancellation Rate

=

Cancelled Trips

Total Valid Trips

But only consider trips where:

Client is NOT banned
AND
Driver is NOT banned

and

Date between

2013-10-01
And
2013-10-03

Query Query History

```sql
INSERT INTO Trips
VALUES
(1,1,10,1,'completed','2013-10-01'),
(2,2,11,1,'cancelled_by_driver','2013-10-01'),
(3,3,12,6,'completed','2013-10-01'),
(4,4,13,6,'cancelled_by_client','2013-10-01'),
(5,1,10,1,'completed','2013-10-02'),
(6,2,11,6,'completed','2013-10-02'),
(7,3,12,6,'completed','2013-10-02'),
(8,2,12,12,'completed','2013-10-03'),
(9,3,10,12,'completed','2013-10-03'),
(10,4,13,12,'cancelled_by_driver','2013-10-03');

Select request_at as day, round(avg(case when t.status='cancelled_by_client' or t.status='cancelled_by_driver'
then 1 else 0 end),2) cancellation_rate
from Trips t
left join Users u
on t.client_id=u.users_id
and u.banned='No'
left join Users u1 on
t.driver_id=u1.users_id
and u1.banned='No'
where request_at between '2013-10-01' and '2013-10-03'
group by day
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	day date	cancellation_rate numeric
1	2013-10-01	0.50
2	2013-10-02	0.00
3	2013-10-03	0.33